Prabhavit Gupta



Deployed Engineer: Agent Architectures, Orchestration & Enterprise Integrations

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Summary

AI Engineer and Lead Developer with 3 years of hands-on production experience architecting autonomous agent systems, multi-step orchestration workflows, and data-intensive enterprise integrations. Deep practical expertise guiding complex enterprise use cases across the full Agent Development Lifecycle (ADLC)—translating ambiguous requirements into rigorous software specifications, stateful agent graphs, schema-enforced tool execution, and deterministic validation loops. Experienced building with LangChain, LangGraph state patterns, orchestration runtimes (n8n, custom engines), and Python/TypeScript microservices. Specialized in evaluation harnesses analyzing real execution traces, intent routing, latency profiling, and human-in-the-loop escalation. Proven track record co-building alongside technical stakeholders from discovery to production rollout. Holder of an EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all member states without visa sponsorship); based in Venice, Italy and available for immediate deployment and up to 20% travel.

Technical Expertise

- **Agent Engineering & ADLC:** Autonomous multi-step agents, LangChain, LangGraph state machine patterns, schema-enforced tool calling, deterministic validation loops, intent routing, memory persistence, human-in-the-loop fallback escalation.
- **Evaluation & Observability:** Trace analysis over production transcripts, structured eval harnesses for non-deterministic AI systems, error logging and root-cause debugging, latency and API token profiling.
- **Workflow Orchestration & Tooling:** Advanced n8n pipelines, custom integration nodes, Model Context Protocol (MCP) servers, Claude Code agentic workflows, asynchronous queues (Redis, Celery).
- **Integration & Backend Architecture:** Python (FastAPI), TypeScript, Node.js, Next.js, PostgreSQL, Redis, RESTful APIs, GraphQL, webhook event architectures, OAuth2/OIDC, JSON Schema validation.
- **Client-Facing Advisory & Delivery:** Technical discovery, architectural reviews, proof-of-concept/value delivery, customer co-building, technical documentation, cross-functional collaboration with engineering leaders.
- **Languages & Authorization:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect *London, UK (remote)*

- Architected, deployed, and maintained the production AI customer support and workflow automation runtime serving 4,500+ active enterprise users across pricing, venue inventory, and multi-location bookings.
- Grounded AI agent responses against live operational databases and partner APIs (OfficeRnD, Nexudus) using strict JSON tool schemas and deterministic validation, eliminating hallucinated answers.
- Built automated evaluation loops analyzing real customer conversation transcripts and execution traces rather than synthetic benchmarks, detecting failure patterns and establishing deterministic fallback escalation paths.
- Engineered resilient webhook listeners and event-driven data synchronization pipelines connecting platform datastores with external CRM systems and scheduling tools, maintaining 99.9% uptime.
- Profiled latency and API token consumption across model versions, optimizing prompt templates and context window utilization to reduce inference costs while maintaining sub-second response times.
- Deployed automated end-to-end regression test suites using Playwright, verifying API contracts and continuous deployment stability across end-to-end customer booking flows.
- Spearheaded platform migration from legacy architecture to a modern Next.js and backend service stack, supporting platform growth from 2,000 to 4,500+ users without operational degradation.

Urlink **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Managed client discovery and solution architecture for bespoke web applications, defining technical specifications, API contracts, and integration milestones.

- Built automated validation and data synchronization pipelines connecting third-party services to PostgreSQL datastores, eliminating manual entry and ensuring strict relational integrity.
- Established CI/CD deployment routines and automated quality checks to accelerate delivery cycles for client deliverables.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed and delivered tailored automation workflows, web data extraction engines, and API integrations for European SME clients using Python, TypeScript, and modern orchestration tools (n8n, Make).
- Replaced fragmented, manual spreadsheet operations with robust event-driven integrations featuring automatic retry queues, dead-letter alerts, and structured audit logs.
- Conducted hands-on technical onboarding, client stakeholder demos, and workflow training, achieving high user adoption across newly automated operational processes.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous grounding in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.